

第二章 几何光学成像

§1 成像

1.1 同心光束：本身或延长线交于一点

$\left\{ \begin{array}{l} \text{实物：入射同心光束是会聚的} \\ \text{虚物：—— 发散的} \end{array} \right.$

1.2 共轭性 $\sim$ 共轭点

1.3

光程：

Q



光程

Q'

$$L_{Q,Q'} = -n \cdot \overline{Qm} + \overline{mQ'}$$

(L2) 物与像折射率

1.4 等光程角

(1) 反射 $\rightarrow$ 椭圆、抛物线、双曲线、平面

(2) 折射 $\rightarrow$ 笛卡儿圆

特殊：齐明点

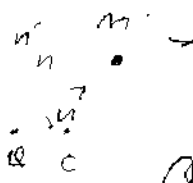
$$\overline{QC} = \frac{n'}{n} r, \overline{Q'C} = \frac{n}{n'} r \rightarrow \triangle QnC \sim \triangle mQ'C$$

$$\frac{\overline{Qm}}{\overline{mQ'}} = \frac{\overline{mC}}{\overline{Q'C}} = \frac{n}{n'}$$

$$\Rightarrow L_{Q,Q'} = n \overline{Qm} - n' \overline{Q'm} = 0$$

$$* RQ = r \left(1 + \frac{n}{n'}\right) = \frac{1}{n'} \cdot (n' + n) r \quad RQ' = r \left(1 + \frac{n'}{n}\right) = \frac{1}{n} (n + n') r$$

$$\frac{\sin u}{\sin u'} = \frac{\overline{QC}}{\overline{mC}} = \frac{n'}{n} = \frac{RQ}{RQ'}$$



§ 2. 共轴球面镜成像

2.1 单球面折射

$$n \sin i = n' \sin i'$$

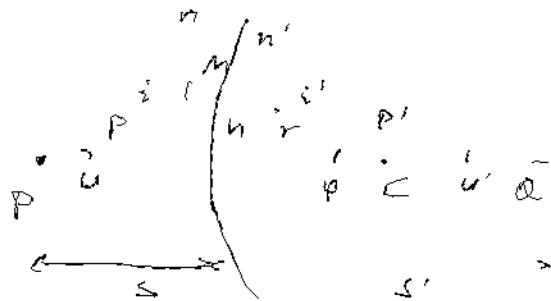
$$i - u = i' + u' = \varphi$$

$$\frac{p}{s \sin \varphi} = \frac{r}{s \sin u} = \frac{s+r}{s \sin i}$$

$$p^2 = r^2 + (s+r)^2 - 2r(s+r) \cos \varphi =$$

$$\frac{p'}{s' \sin \varphi} = \frac{r}{s' \sin u'} = \frac{s'-r}{s' \sin i'}$$

$$p'^2 = r^2 + (s'-r)^2 + 2r(s'-r) \cos \varphi$$



$$\Rightarrow \frac{p}{(s+r)n} = \frac{p'}{(s'-r)n'}$$

$$p^2 = (s+r-r)^2 + 2(s+r)r(1-\cos \varphi) = s^2 + 4(s+r)r \sin^2(\frac{\varphi}{2})$$

$$p'^2 = s'^2 - 4(s'-r)r \sin^2(\frac{\varphi}{2})$$

$$\Rightarrow (s'-r)^2 n'^2 \cdot [s^2 + 4(s+r)r \sin^2(\frac{\varphi}{2})] = (s+r)^2 n^2 \cdot [s'^2 - 4(s'-r)r \sin^2(\frac{\varphi}{2})]$$

$$n'^2(s'-r)^2 \cdot s^2 - n^2(s+r)^2 s'^2 = -4rs \sin^2(\frac{\varphi}{2}) (s'-r)(s+r) [n'^2(s+r) + n^2(s'-r)]$$

$$\text{约去 } n^2 \cdot n'^2, s+r, s'-r$$

$$\frac{s^2}{n^2(s+r)^2} - \frac{s'^2}{n'^2(s'-r)^2} = -4rs \sin^2(\frac{\varphi}{2}) \left[\frac{1}{n'^2(s'-r)} + \frac{1}{n^2(s+r)} \right]$$

① 由 φ 由成 $\Rightarrow$ 齐明点

2.2 轴上物点成像

② $\varphi \ll 1$ 即 $h \ll s, s', r$ / $u, u', \varphi \ll 1$ 傍轴条件

$$\Rightarrow \frac{s^2}{n^2(s+r)^2} = \frac{s'^2}{n'^2(s'-r)^2}$$

$$\Rightarrow \frac{n}{s+r} = \frac{n'}{s'-r} = n' - n$$

焦距: $s' \rightarrow \infty, f = \frac{nr}{n'-n}$

$s \rightarrow \infty, f' = \frac{n'r}{n'-n}$

Gauss 公式: $\frac{f}{s} + \frac{f'}{s'} = 1$

反射: 取 $n' = -n \rightarrow \frac{1}{s} - \frac{1}{s'} = \frac{2}{r}$

符号规则:

实/虚物: $\begin{cases} Q \text{ 在 } A \text{ 左}, s > 0 \\ Q \text{ 在 } A \text{ 右}, s < 0 \end{cases}$

实/虚像: 折 $\begin{cases} Q' \text{ 在 } A \text{ 右}, s' > 0 \\ Q' \text{ 在 } A \text{ 左}, s' < 0 \end{cases}$ 反 $\begin{cases} Q' \text{ 在 } A \text{ 左}, s' > 0 \\ Q' \text{ 在 } A \text{ 右}, s' < 0 \end{cases}$

半径: $\begin{cases} C \text{ 在 } A \text{ 左 (左凹球面)}, r < 0 \\ C \text{ 在 } A \text{ 右 (右凹球面)}, r > 0 \end{cases}$

像: $\begin{cases} \text{正立} & y' > 0 \\ \text{倒立} & y' < 0 \end{cases}$

角度: $\begin{cases} \text{逆时针旋转}, u > 0 \\ \text{顺时针旋转}, u < 0 \end{cases}$

x/x' : $\begin{cases} Q \text{ 在 } F \text{ 左}, x > 0 & Q \text{ 在 } F \text{ 右}, x < 0 \\ Q' \text{ 在 } F \text{ 右}, x' > 0 & Q' \text{ 在 } F \text{ 左}, x' < 0 \end{cases}$

2.3 傍轴物点成像

OP 物平面, π 像平面, π' ~ 共轭平面

(2) ① 横向放大率 $V = \frac{y'}{y}$

$$i, i' \ll 1$$

$$ni = n'i'$$

$$i \approx \frac{y}{s}, \quad i' \approx -\frac{y'}{s'}$$

$$\Rightarrow V = -\frac{ns'}{n's}$$

$$\hookrightarrow \text{反射: } V = -\frac{s'}{s}$$



② 角放大率 W

$$W = \frac{\tan u'}{\tan u} \quad u = \frac{h}{s}$$

2.4 逐次成像

2.5 L-H 定理

对于一次折射:

$$u = \frac{h}{s}, \quad u' = -\frac{h}{s'}$$

$$\Rightarrow \frac{u'}{u} = -\frac{s}{s'} = \frac{n}{n'V} = \frac{ny}{n'y'}$$

$$\Rightarrow ynu = \text{const.}$$

f = 薄透镜

3.1 成像公式

$$\text{前: } \frac{f_1}{s_1} + \frac{f_1'}{s_1'} = 1$$

$$\text{或: } \frac{n}{s_1} + \frac{n_L}{s_1'} = \frac{n_L - 1}{r_1}$$

$$\text{后: } \frac{f_2}{s_2} + \frac{f_2'}{s_2'} = 1$$

$$\frac{n_L}{-s_1'} + \frac{n'}{s_2'} = \frac{1 - n_L}{-r_2}$$

$$s_2 = -s_1' + \underbrace{d}_{\approx 0} \approx -s_1'$$

$$\Rightarrow \frac{n}{s} + \frac{n'}{s'} = (n_L - 1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$\Rightarrow \frac{f_1 f_2}{s} + \frac{f_1' f_2'}{s'} = f_2 + f_1'$$

$$s' \rightarrow \infty, f = \frac{f_1 f_2}{f_2 + f_1'} = \frac{n}{(n_L - 1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)} = n f_0$$

$$s \rightarrow \infty, f' = \frac{f_1' f_2'}{f_1 + f_2} = \frac{n'}{(n_L - 1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)} = n' f_0$$

$$\text{取 } n = n' = 1 \Rightarrow \text{薄透镜公式: } \frac{1}{f} = \frac{1}{(n_L - 1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)}$$

3.2 成像公式

$$\frac{f}{s} + \frac{f'}{s'} = 1 \quad n = n' = 1 \quad \Rightarrow \quad \frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

$$\text{又: } x = s - f, x' = s' - f'$$

$$\rightarrow x x' = f f'$$

$$V = V_1 \cdot V_2 = \left(-\frac{n s_1'}{n_L s_1} \right) \left(-\frac{n_L s_2'}{n' s_2} \right) = -\frac{n s_1'}{n' s_1} \quad (s_1' = -s_2)$$

$$= -\frac{n s'}{n' s}$$

$$= -\frac{f s'}{f' s} \quad \left(f = \frac{n r}{n_L - n}, f' = \frac{n' r}{n' - n} \right)$$

$$v = -\frac{f}{x} = -\frac{x'}{f'} \quad \therefore -\frac{f s'}{f' s} = -\frac{f(x'+f')}{f'(x+f')} = -\frac{f(x'+f')}{f'x+xx'} = -\frac{f}{x} = -\frac{x'}{f'}$$

3. 先焦距 p

复合透镜 / 密接透镜

$$\frac{1}{s_1} + \frac{1}{s_1'} = \frac{1}{f_1} \quad , \quad \frac{1}{s_2} + \frac{1}{s_2'} = \frac{1}{f_2}$$

$$\frac{1}{s_1} + \frac{1}{s_2'} = \frac{1}{f}$$

$$s_1' = -s_2$$

$$\Rightarrow \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\text{定义: } P = \frac{1}{f} \quad \rightarrow \quad P = P_1 + P_2$$

例 1: 设 L_1 和 L_2 , $f_1 = 20 \text{ cm}$, $f_2 = 40 \text{ cm}$, L_2 在 L_1 右 40 cm 处

紧贴小物在 L_1 左 30 cm , 求像

$$\frac{1}{s_1} + \frac{1}{s_1'} = \frac{1}{f_1} \quad \rightarrow \quad s_1' = 60 \text{ cm}, \quad V_1 = -\frac{s_1'}{s_1} = -2$$

$$-(s_1' - d) + \frac{1}{s_2} = -\frac{1}{f_2} \quad \rightarrow \quad s_2' = 40 \text{ cm}, \quad V_2 = -\frac{s_2'}{d - s_1'} = 2$$

$$V = V_1 \cdot V_2 = -4$$

§4 理想光学系统理论

4.1 理想成像和共轭变换

空间中任一点都严格成像 / 同心光束 $\rightarrow$ 同心光束

性质: ① 共轭变换

物方一点 / 线 / 平面

$\Leftrightarrow$ 像方一点 / 线 / 平面

(轴对称时) ② 光轴一点 $\rightarrow$ 光轴一点

③ 与轴垂直平面 $\rightarrow$ 与轴垂直平面

④ 同一平面内 $\vee$ 相同 $\rightarrow$ 几何相似性

望远系统 ⑤ 两垂直于光轴平面 $\vee$ 相同 ($V_{n_1 n_1'} = V_{n_2 n_2'}$)

$\rightarrow$ 处处 $\vee$ 相同

4.2 基点 / 基面

① 焦点 F / 焦面 Σ

$\rightarrow$ 与无穷远平面共轭

② 主点 H / 主面 Σ_H

$\rightarrow$ 横向放大率为1的共轭面。

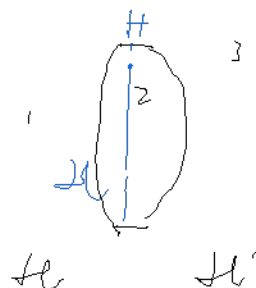
Q: 为什么 H 是 1, 3 点?

A: 1 要过 H , 与 H 交于 H , 3 要过 H' , 交 H' 于 H'

$$y_{H'} = |V| \cdot y_H = y_H$$

$$\tan u'$$

$$= \frac{y}{f}$$



$\tan u = \frac{y}{x} = \frac{y'}{x'} \Rightarrow V = \frac{y'}{y} \Rightarrow \frac{y'}{y} \cdot \frac{\tan u'}{\tan u} = \frac{n}{n'}$

$\Rightarrow y \tan u = y' n' \tan u' = \text{const.}$

3) 节点 N

↳ 轴上角放大率为1的点

4.3 理想光具座的联合

$$\frac{y}{n} = \frac{f'}{\Delta + f_2} = \frac{-f'}{f_2' + x_2'} \Rightarrow f' = - \frac{f_1'(f_2' + x_2')}{\Delta + f_2}$$

F_1', F' 共轭 $\Rightarrow$ 牛顿公式 $\Delta x_2' = f_2 f_2'$

$\Rightarrow f' = - \frac{f_1'(f_2' + \frac{f_2 f_2'}{\Delta})}{\Delta + f_2} = - \frac{f_1' f_2'}{\Delta}$

$x_{H'} = f_2' + x_2' - f' = f_2' + \frac{f_2 f_2'}{\Delta} + \frac{f_1' f_2'}{\Delta}$

$= f_2' \left(\frac{f_2 + f_1'}{\Delta} + 1 \right) = f_2' \frac{d}{\Delta} = f_2' \cdot \frac{d}{-f_1' f_2'} \Rightarrow f = - \frac{f_1' d}{f_1'}$

$(d = f_1' + \Delta + f_2)$

↓ 由对称性

$f = - \frac{f_1' f_2'}{\Delta}$

$P = \frac{1}{f}, n = n' = 1 (f = f')$

$P = P_1 + P_2 - P_1 P_2 d$

$$X_{14} = f_1 f_2 (f_1 + f_2 + 1) = d - f_1 - f_2$$

$$= f_1 d = - \frac{f d}{f_2}$$

§ 5. 光学仪器

1. 放大镜

明视距离: $s_0 = 25 \text{ cm}$

最大视角 $\omega = \frac{y}{s_0}$

放大镜: 成像在 $s_0 \sim \infty$ 之间

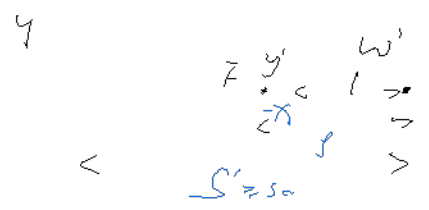
$$x x' = f \cdot f' = f^2$$

$$\Rightarrow -x = \frac{f^2}{x'} \in (0, \frac{f^2}{s_0 + f})$$

焦距 $\Delta x = \frac{f^2}{s_0 + f} \approx \frac{f^2}{s_0} \quad (f \ll s_0)$

视角: $\omega' \approx \frac{y}{f}$

视角放大率: $M = \frac{\omega'}{\omega} = \frac{s_0}{f}$



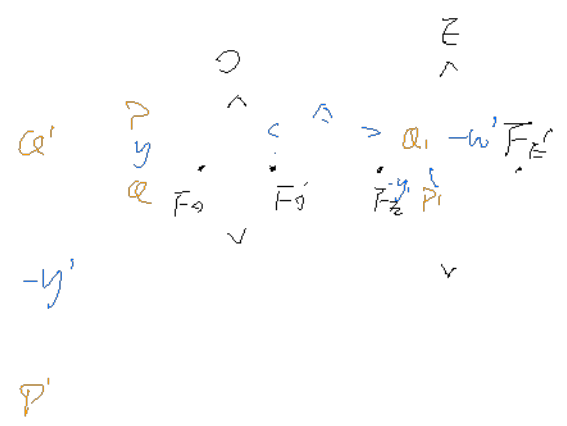
2. 望远镜

$M = \frac{\omega'}{\omega} = \frac{y_1}{f_2} \cdot \frac{s_0}{f_1}$ (成像在 F_2 附近)

$= \frac{y_1}{y} \cdot \frac{s_0}{f_2} = V_0 \cdot M_2$ (V_0 为物镜横向放大率, M_2 为目镜视角放大率)

$$V_0 = -\frac{f}{x} = -\frac{x'}{f_1} = -\frac{\Delta}{f_0}$$

其中 Δ 为光筒长

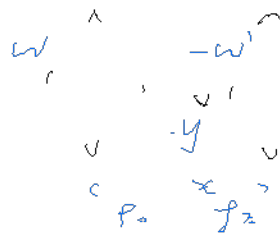


$$f_0 f_E$$

3. 望远镜

物镜成像在焦点附近， F_0' 与 F_E 几乎重合

$$M = \frac{w'}{w} = \frac{-y}{-y} = -\frac{f_0}{f_E}$$



4. 棱镜

$$\text{角色散本领 } D = \frac{d\delta_{min}}{d\lambda}$$

$$n = \frac{\sin(\frac{\alpha + \delta_{min}}{2})}{\sin(\frac{\alpha}{2})}$$

$$= \frac{d\delta_{min}}{dn} \cdot \frac{dn}{d\lambda}$$

$$\frac{d\delta_{min}}{dn} = \frac{2\sin(\frac{\alpha}{2})}{\cos(\frac{\alpha + \delta_{min}}{2})} = \frac{2\sin(\frac{\alpha}{2})}{1 - n^2 \sin^2(\frac{\alpha}{2})}$$

$$= \frac{2\sin(\frac{\alpha}{2})}{1 - n^2 \sin^2(\frac{\alpha}{2})} \cdot \frac{dn}{d\lambda}$$

$$\propto \frac{dn}{d\lambda} \quad \text{色散率}$$